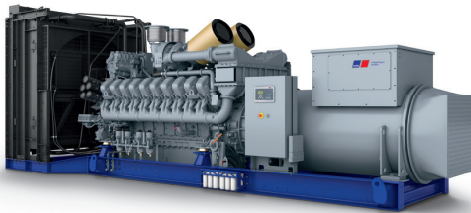




Diesel Generator Set

mtu 20V4000 DS4000

400 V - 11 kV/50 Hz/standby power/fuel consumption optimized
20V4000G94LF/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits

- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support

- Global product support offered

Standards

- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating

- System ratings: 3950 kVA - 4000 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)

- Engine-generator set tested to ISO 8528-5 for transient response
- 85% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available

- Control panel
- Power panel
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Electrical driven radiators
- Medium and oversized voltage alternators
- Low voltage alternator

Emissions

- Fuel consumption optimized

Certifications

- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary



A Rolls-Royce
solution

Application data ¹⁾

Engine			Liquid capacity (lubrication)	
Manufacturer		<i>mtu</i>	Total oil system capacity: l	390
Model	20V4000G94LF		Engine jacket water capacity: l	260
Type	4-cycle		Intercooler coolant capacity: l	50
Arrangement	20V		Combustion air requirements	
Displacement: l	95.4		Combustion air volume: m³/s	4.5
Bore: mm	170		Max. air intake restriction: mbar	30
Stroke: mm	210		Cooling/radiator system	
Compression ratio	16.4		Coolant flow rate (HT circuit): m³/hr	80
Rated speed: rpm	1500		Coolant flow rate (LT circuit): m³/hr	44
Engine governor	ADEC (ECU 9)		Heat rejection to coolant: kW	1220
Max power: kWm	3308		Heat radiated to charge air cooling: kW	840
Air cleaner	dry		Heat radiated to ambient: kW	105
Fuel system			Fan power for electr. radiator (40°C): kW	105
Maximum fuel lift: m	5		Exhaust system	
Total fuel flow: l/min	27		Exhaust gas temp. (after engine): °C	481
Fuel consumption ²⁾			Exhaust gas temp. (before turbocharger): °C	693
At 100% of power rating:	l/hr	g/kwh	Exhaust gas volume: m³/s	11.5
At 75% of power rating:	806	202	Maximum allowable back pressure: mbar	50
At 50% of power rating:	565	189	Minimum allowable back pressure: mbar	-
	403	202		

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	Fuel consumption optimized 40°C/300m		
		without radiator		
		kWel	kVA*	AMPS
Leroy Somer LSA54.2 ZL17 (LV Leroy Somer standard)	400 V	3200	4000	5774
Leroy Somer LSA54.2 ZL12 (Medium volt. Leroy Somer)	11 kV	3160	3950	207
Marathon 1040FDH7105 (Medium volt. marathon)	11 kV	3200	4000	210
Leroy Somer LSA54.2 ZL14 (MV Leroy Somer oversized)	11 kV	3160	3950	207
Leroy Somer LSA54.2 ZL14 (Engine output optimized)	11 kV	3200	4000	210

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).
2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.